

# VICTORIA TJHIN

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## EDUCATION

- University of Waterloo - BASc, Faculty of Engineering, Honors in Nanotechnology Engineering** | 2024 - 2029 (exp.)
- GPA: 3.92/4.00, awarded Consecutive Term Distinction
  - Chemical Engineering (2024 - 2025)

## TECHNICAL SKILLS

**Languages & HDL** Verilog, SystemVerilog, C++, Python, MATLAB, SQL

**EDA & Tools** Xschem, Magic VLSI, Netgen, Ngspice, Sky130 PDK, Vivado, GTKWave, Tiny Tapeout, KiCad

## PROJECTS

**ASIC Optical Autoalignment System - SSCS 2026 Chipathon** | Mixed-Signal ASIC, RTL, DSP

- Engineered analog integrator and mixed-signal front end converting nanoamp photodetector current into signal alignment error signals.
- Architected parallel X/Y RTL DSP pipeline closing the 2D position loop, correcting alignment offset every dither cycle at 50 MHz.
- Restructured handshaking across analog, DSP, and actuator blocks, reducing latency 40% and clearing the block for tape-out.

**Ethernet MAC Receiver Pipeline - UWASIC** | SystemVerilog, Digital Design, Verification

- Developed a Verilog MAC receive datapath from PCS decoder, preamble/SFD detection, FCS check, and AXI-Stream output.
- Pipelined CRC-32 engine in SystemVerilog with keep-aware partial-beat handling, carrying 64-bit beats at 10 Gb/s.
- Implemented asynchronous FIFO system clock domain crossing between PCS and system clock domains with randomized testbenches catching CRC defects.

**SPI-Controlled PWM Peripheral - UWASIC** | Verilog, Tiny Tapeout Submission

- Implemented a register mapped SPI slave and 8-bit PWM generator in Verilog with duty, output-enable, and control logic.
- Validated 256 duty cycle settings across 100 kHz SPI bus and 10 MHz core clock, verifying pulse width and bus integrity.

**Two-Stage CMOS Operational Amplifier - UWASIC** | Sky130 130 nm, Analog IC Design

- Designed a differential input, two-stage CMOS op-amp achieving >60 dB gain within a 5 mW power budget.
- Sized transistors from calculations, validated gain, bandwidth and operating points through DC and AC simulation in Ngspice.
- Completed physical layout in Magic with DRC-clean and LVS-matched results, clearing design for fabrication.

**Maskless Lithography Machine - HackerFab** | KiCAD and Thermal PCB

- Engineered the LED illumination and thermal PCB in KiCad, integrating drivers, thermistor feedback, and power delivery.
- Analyzed heat dissipation and airflow to maintain LED temperature within 5 °C stabilizing photoresist dose.
- Assembled, tested, and calibrated the board through bring-up into a working exposure subsystem.

**Conductive Hydrodynamic Ink Printer - UWNRG** | C++, Embedded, PCB

- Engineered stepper driver PCBs in KiCad, validating firmware-to-motor signalling across a three-axis motion system.
- Developed Marlin firmware in C++ synchronizing three-axes and lead screw logic within 50 µm of target.
- Modeled the cartridge mount and lead screw drive in SolidWorks, reducing motion system backlash by 30%.

## WORK EXPERIENCE

**Institute for Quantum Computing - Research Assistant** Jan 2025 - Present | Waterloo, Canada

- Fabricated CNT-based single photon detectors through chemical functionalization, drop-casting, and roll-coating.
- Characterized and Identified CNT deposition configuration using UV-Vis, Raman, and SEM, increasing device yield by 25%.
- Automated UV-Vis spectral analysis in Python across 200+ runs, cutting processing time by 90%.

**UW Chemical Engineering - Research Assistant** Sep - Dec 2025 | Waterloo, Canada

- Designed a graphene exfoliation setup in SolidWorks and prepared the components for fabrication in Cura
- Synthesized graphene by electrochemical exfoliation, filtration, and sonication achieving 99.9% purity, 87% yield; and Electrospun nanofibers, with fire retardant and superhydrophobic properties above 150 °C.

**Ontario Battery & Electrochemistry Centre - Battery Intern** May - Aug 2026 | Waterloo, Canada

- Prototyped 40+ Li-ion coin and pouch cells end to end, from slurry prep through coating, tab welding, assembly.
- Characterized 5 cell build using Maccor cycling and impedance spectroscopy, quantifying resistance and C-rate for each.
- Correlated cell variance to electrode porosity, calendaring pressure, and coating temperature, improving consistency 20%.

**Scotiabank - Workplace Transformation Analyst** Jan - Apr 2025 | Toronto, Canada

- Analyzed 2 years of badge and booking data across 6 ecosystems in SQL, identifying 23 workspaces below 40% utilization.
- Translated utilization analysis into retrofit proposal, securing executive approval to redesign all 23 spaces.
- Presented ad-hoc analysis to executive leadership, supporting sustainment strategy for 13,000+ employees.